an edge beyond which spacetime ceases to exist)?" (Figure 13.5c) and "Which regions of spacetime can send signals to each other, and which cannot?" (Figure 13.5d). The first of these topological questions is central to singularities; the second is central to the formation and existence of black holes and also to *cosmology* (to the large-scale structure and evolution of the Universe).

These topological issues are so important, and the mathematical tools of topology are so powerful in dealing with them, that by introducing us to topology, Penrose triggered a revolution in our research.

Taking off from Penrose's seminal ideas, during the middle and late 1960s Penrose, Hawking, Robert Geroch, George Ellis, and other physicists created a powerful set of combined topological and geometrical tools for general relativity calculations, tools that are now called *global methods.* Using these methods, Hawking and Penrose in 1970 proved—without any idealizing assumptions-that our Universe must have had a spacetime singularity at the beginning of its big bang expansion, and if it one day recollapses, it must produce a singularity in its big crunch. And using these global methods, Hawking in 1970 invented the concept of a black hole's absolute horizon and proved that the surface areas of absolute horizons always increase (Chapter 12).

Let us return, now, to 1965. The stage was set for a momentous confrontation. Isaac Khalatnikov and Evgeny Lifshitz in Moscow had proved (or so they thought) that when a real star, with random internal deformations, implodes to form a black hole it *cannot* create a singularity at the hole's center, while Roger Penrose in England had proved that every black hole *must* have a singularity at its center.